The Slide Rule

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1 First the nitty gritty

To understand the action of a Slide Rule we have to start from Logarithms, or logs for short.

Consider the two equations:

$$x = b^y$$

$$y = \log_b x$$

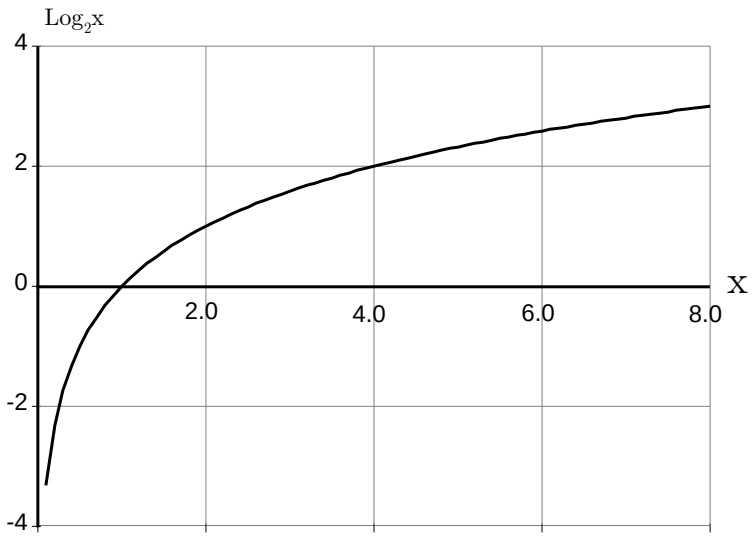
These say the same thing, one gives x in terms of y (and b)
The other gets y in terms of x (and b).

In this case b is called the base of the log.
So an example:

$$8 = 2^3$$

$$3 = \log_2 8$$

To illustrate this, here's a graph of $\log_2 x$ against x.



A point to notice is that no negative x values are plotted. Logarithms are not defined for negative x, but the logs themselves do take negative values as x becomes less than 1.0

$$0.5 = 2^{-1}$$

$$-1 = \log_2 0.5$$

1.1 The heart of the matter

Given $y = \log_b x$ and $x = b^y$ we have an identity we are going to use later:

$$y = \log_b b^y$$

Here's the magic, take two numbers $c = b^p$ and $d = b^q$, so therefore:

$$p = \log_b c$$

$$q = \log_b d$$

We know

$$cd = b^p b^q = b^{p+q}$$

To illustrate:

$$8 \times 4 = 2^3 \times 2^2 = 2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$$

Taking the log of both sides, and noting that $y = \log_b b^y$

$$\log_b cd = \log b^p b^q = \log_b b^{p+q} = p + q$$

But we know $p = \log_b c$ and $q = \log_b d$ so lets go all the way

$$\log_b cd = \log_b c + \log_b d$$

So that means we are mapping the multiplication of two numbers, to an addition! Which is the basis of how a slide rule works.

The problem seems to be twofold, we need to find the logs of any two numbers, and after adding them, we get the log of their product - so we need to get back to the desired product from its log - that is, we need to get the anti-log.

So how do we get logs and anti-logs? There are techniques such as power series that converge on the logarithm of a value, however for this article, we don't bother, a slide rule has logarithms built-in, the length along a slide rule is the log_{10} of the number inscribed on the rule. More of this later.

As a school child, everyone over a certain age had a booklet of 'Logarithmic Tables' which also included anti-log and trigonometric 'four-figure' tables. Ask your granddad.

To continue, how about division:

$$\frac{c}{d} = \frac{b^p}{b^q} = b^p b^{-q} = b^{p-q}$$

And that leads to

$$\log_b \frac{c}{d} = \log_b b^{p-q} = p - q = \log_b c - \log_b d$$

So division maps to subtraction, that's neat.

Now let's look at another sweet property. Starting again from:

$$x = b^y$$

$$x^n = b^{ny}$$

Take the log of both sides:

$$\log_b x^n = \log_b b^{ny}$$

$$\log_b x^n = ny$$

$$\log_b x^n = n \log_b x$$

We've reduced finding a power to a multiplication! The schoolboy would have to look up the log of a number x, do the n multiplication, and then look up the anti-log to get the final result x^n, and that's a powerful technique. As n can be fractional, this includes roots as well.

And if we really want to push the boat out, we can do that n times multiplication with logs:

$$\log_a(\log_b x^n) = \log_a(n \log_b x) = \log_a n + \log_a(\log_b x)$$

I've introduced another base a here, just to show that the multiplication is not necessarily in base b again.

So there's a few things to notice:

- $\log_b 1 = 0$ for any base, since $b^0 = 1$
- $\log_b b = 1$ for any base, since $b^1 = b$
- $\log_b x^n = n \log_b x$
- $\log_b(\frac{1}{x})$=-$\log_b x$

If you check the earlier graph, its worth spotting these points in action. For further information about logarithms, a good description is given at [1]

1.2 All about bases

Let's dive a bit deeper into that base number b, especially with respect to Slide Rules. First we'll show that changing bases is surprisingly easy.

Starting from these two again:

$$y = \log_b x$$

$$x = b^y$$

Take the log (with base c) of both sides

$$\log_c x = \log_c b^y$$

$$\log_c x = y \log_c b$$

$$\frac{\log_c x}{\log_c b} = y$$

$$\frac{\log_c x}{\log_c b} = \log_b x$$

So if someone has kindly created a table of logarithms to base c, then creating a new table of logarithms to base b is simply a matter of dividing each value in the table by a constant $\log_c b$

Logs to base e, the exponential constant, are used all over the place by physicists and mathematicians. However the base we are particularly interested in is base 10, and we'll now explore why. From the previous section we now use 10 instead of b:

- $x = 10^y$
- $y = \log_{10} x$
- $\log_{10} 1 = 0$
- $\log_{10} 10 = 1$
- $\log_{10} x^n = n \log_{10} x$
- $\log_{10} 10^n = n \log_{10} 10 = n$

I'll just use log from now on, rather than $\log_{10}$

We see that the values $\log 1$ to $\log 10$ will span the range 0 to 1.

Or put another way, $1 = 10^0$ and $10 = 10^1$ and an intermediate value $5 = 10^{0.7}$(approx).

Given an x value in this range of 1 to 10:

$$0 \leqq \log x \leqq 1$$

So what about $10x$?

$$\log 10x = \log 10 + \log x = 1 + \log x$$

$$1 \leqq \log 10x \leqq 2$$

and it follows

$$3 \leqq \log 100x \leqq 4$$

So in general, if you have calculated the logs of all values between 1 and 10, by simply adding unit values you get all logs for any number greater than 1

The logs of the numbers 10 to 100 are simply one added to the logs of the numbers 1 to 10.

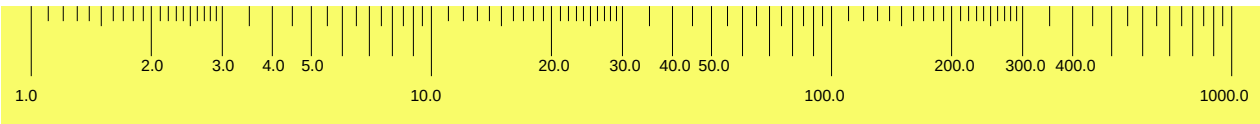
The logs of the numbers 100 to 1000 are simply two added to the logs of the numbers 1 to 10.

It works the other way as well, the logarithms of 0.1 to 1 are one subtracted from the logarithms of the numbers 1 to 10.

The basis of a slide rule is that x values are inscribed, with the distance from the left of the rule to each point being the $\log x$.

The x numbers shown on the rule are therefore the antilogs of the distance along the rule.

To get every number inscribed would take an infinitely long ruler, which could be unwieldy. Below is an image of x values from 1.0 to 1000.0



You might notice there is a level of redundancy here, values from 10.0 to 100.0 do not give any extra information, other than the power of 10, and similarly 100.0 to 1000.0 is just another repeat.

So slide rules can take advantage of this, by just having the numbers 1 to 10 inscribed along the rule.

There is a disadvantage; the slide rule can be used to add (or subtract) the Mantissa (the fractional part) of the logarithm, and ignore the exponent (the units). This means that the result does not give the powers of ten information.

So 9 x 7 may give 63, but the slide rule will not tell you if it is 6.3, or 63, or 630 etc.,

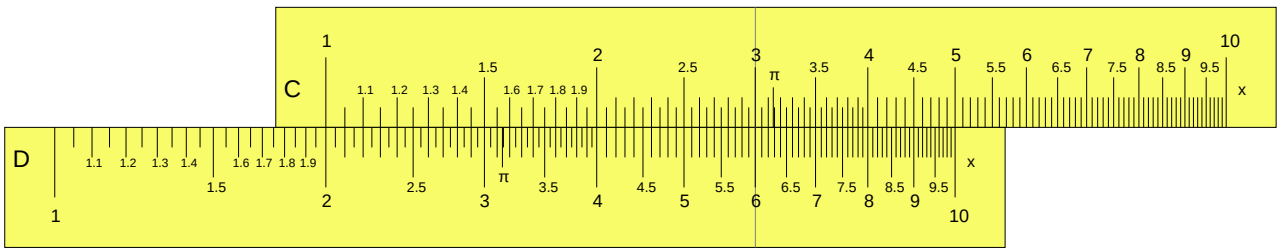
And again for 90 x 7, the slide rule will not tell you if it is 6.3, or 63, or 630 etc.,

So to use a slide rule you also need to use the mark 1 brain to figure out where that decimal point lies.

2 Basic Slide rule usage

2.1 Multiplication

To multiply two numbers using a slide rule, you add the two associated lengths together - the length of a slide rule being the log of the number. The two lengths together give another length (the log of the product), and from that resultant length you can just read off the result. So you need two slide rules, or, as I'm sure you know, the slide rule is made up of sliding parts.



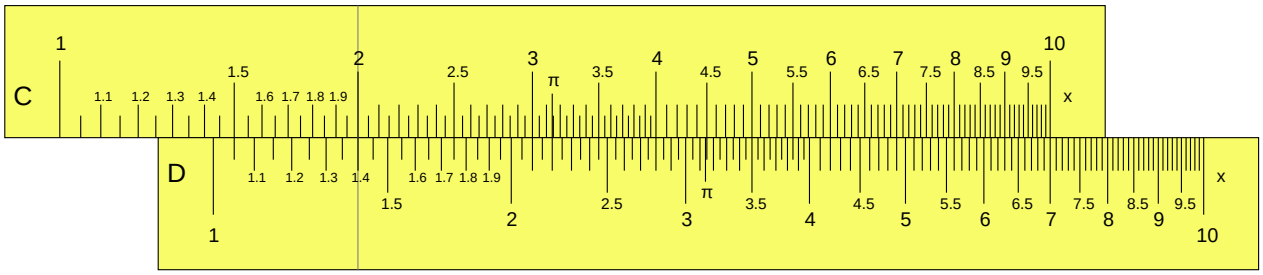
By aligning the 1 of scale C (often called the index) above the 2 of scale D, we have added the $\log 2$ distance to scale C.

Remember that though we have put a '1' over the 2, $\log 1 = 0$, and so in terms of the lengths, this is putting a zero over the 2 position.

By moving along C to any number, say 3, we have added the log of 3. Then reading down onto scale D - which is now the length of $\log 2 + \log 3$,we can see the value of the product, which in this case is 6.

You would do the same thing if you were multiplying 20 by 30. As discussed previously, it is up to you to work out the decimal place.

You may have spotted a problem with this. What if we wanted 2x7? The 7 on the C scale is beyond the end of the D scale. So here's another magic trick:



In this case the 10 of the C scale is positioned over the 7. Then move to the 2 of the C scale, and below it you can see the result 1.4

As the slide rule is indifferent to powers of ten, we have to figure the result is actually 14.

So how are the logarithms being added in this method?

By setting the 10 on to 7, we have the length of the D scale, which is $\log 7$, but then traveling left on the C, from 10 right back to 1, we have subtracted $\log 10$. Then traveling right on the C from 1 to 2, we have added $\log 2$. So we have actually done:

$$\log 7 - \log 10 + \log 2 = \log \frac{7 \times 2}{10}$$

By moving the C scale left we are dividing by 10, but we know not to worry about this, and so we have the significant digits of the result.

In practice, the usage is simple; put either the 1 or the 10 of the sliding C scale over one number, go to the other number on the sliding scale, and read the result on the D scale.

Earlier we mentioned '9 x 7 may give 63', just for fun, on the scale above, you can see the 9 on C is above 6.3 on D.

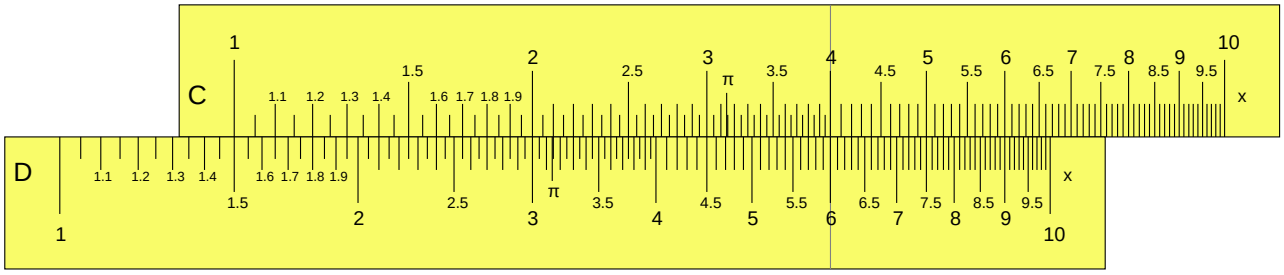
2.2 Division

We know division maps to subtraction with logs.

$$\log \frac{c}{d} = \log c - \log d$$

So how to subtract distance on a slide rule? Lets try six divided by four, which is 1.5

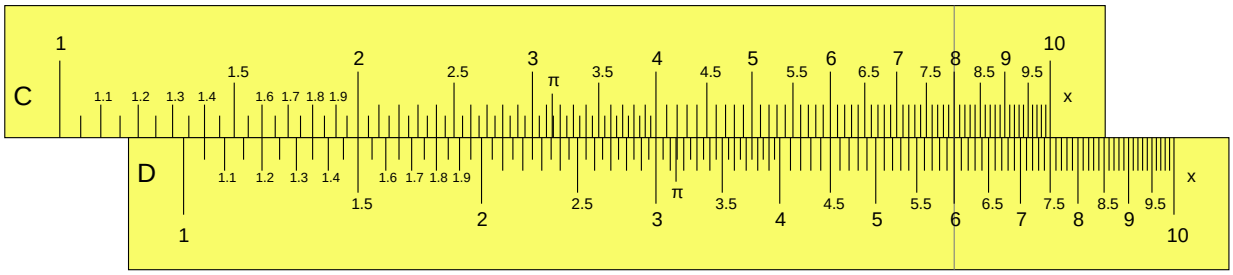
The distance to 6 of the D scale is $\log 6$, move the 4 of the C scale above it, and traveling back along the C scale to the 1 is equivalent to subtracting $\log 4$.



So the distance on the D scale is $\log 6 - \log 4 = \log 1.5$ and by reading the inscribed value, which is the equivalent of reading the antilog, we find 1.5.

Notice, though we see division as $\frac{6}{4}$ the 6 is on the bottom scale, and the four is above it, which is just due to the layout of these sliding scales.

Again we have a problem, what about $\frac{6}{8}$? Moving the 8 above the 6 leaves the 1 on the C scale beyond the D scale.



However the 10 on C scale shows 7.5 on D, and 0.75 is the right answer, so how does that come about?

6 on the D scale is equivalent to $\log 6$, subtracting $\log 8$ takes you leftwards to C1, then adding $\log 10$ takes you to C10.

$$\log 6 - \log 8 + \log 10 = \log \frac{6 \times 10}{8}$$

Powers of ten do not bother us, so reading the antilog on the D scale gives us 7.5, which we realize means 0.75

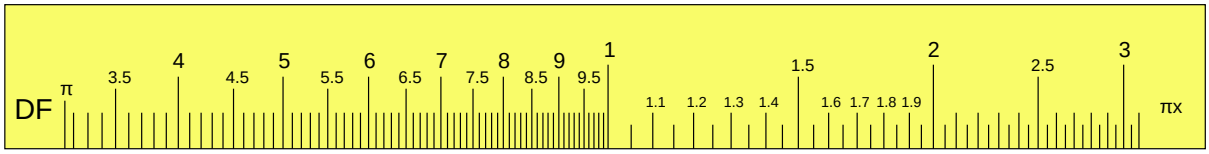
We can be confident that setting a C index, of 1 or 10 over a number and then moving along C is multiplication, and setting a C number over a D number, then moving along C to an index, 1 or 10, gives us a division.

2.3 CF and DF

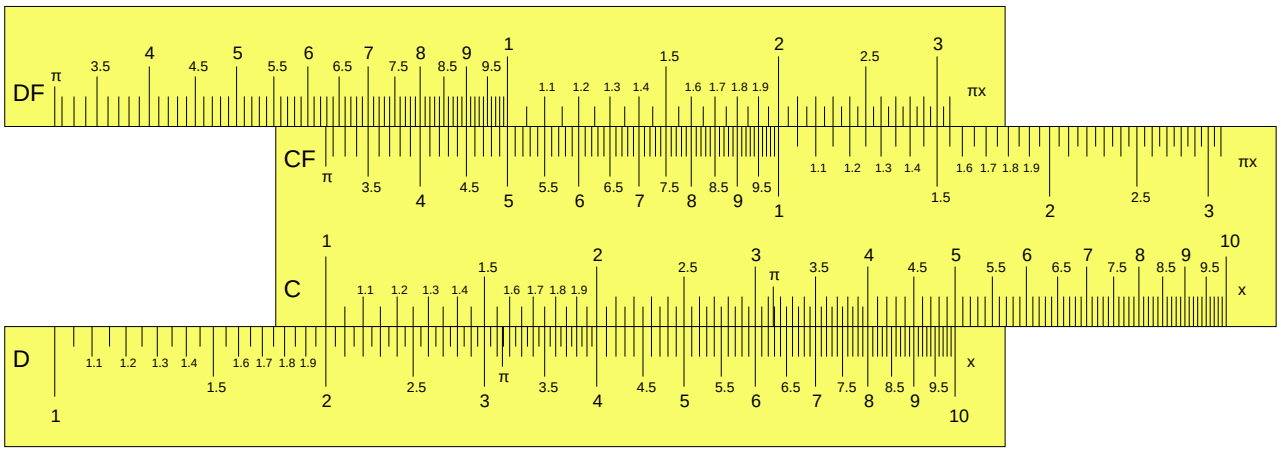
Another twist to the tale of multiplication. Depending on the model, some slide rules have a selection of different scales on both sides which are often useful.

Consider the multiplication of $2 \times 7 \times 3$, using the C10 index we have to change the order to $7 \times 2 \times 3$ which is annoying, especially if your mind is on a calculation rather than how to use a slide rule.

Most Slide Rules also have the scales CF and DF, which are the same as C and D but 'Folded', by the scales being offset by $x = \pi$, and, at what we would consider the 10 point, is labeled as 1. A lone DF shown below:

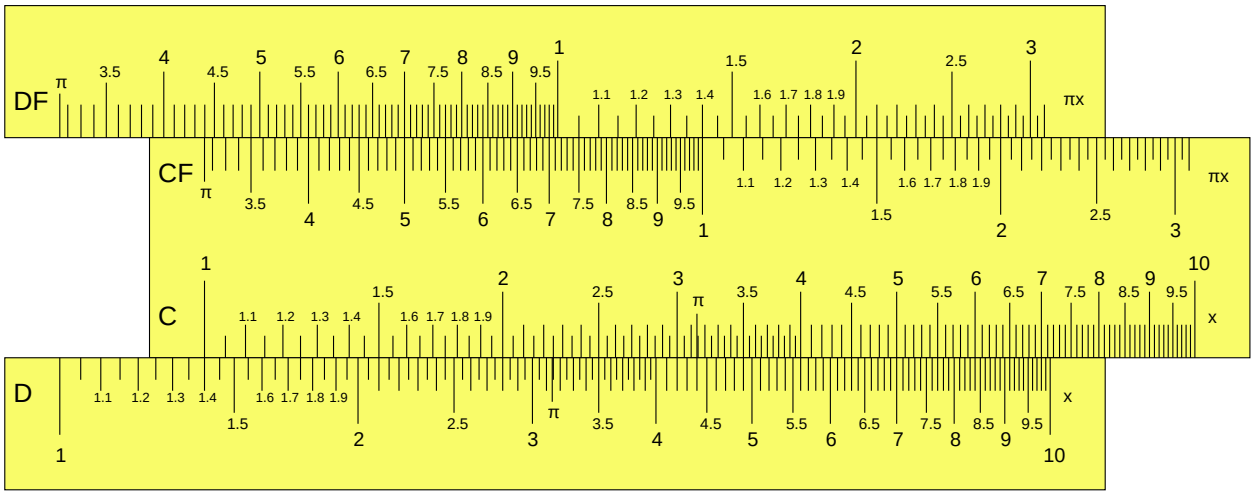


Now look again at 2×7, but using the C1 index, and with a Slide Rule that has CF and DF scales:



With C1 over D2, the C7 is off to the right, but the CF7 is there, and reading on to the DF scale we can see the answer 1.4 (14).

But now we want to multiply 1.4 by three, if we move the CF1 index over 1.4, we find CF3 is off to the right.



But that's still ok, as we can see C3, which is over 4.2 (42), which is the correct answer, we have $2 \times 7 \times 3$, by swapping across the scales.

This works because a movement of the central slider, by $\log 2$ in our initial move, is by a distance of $\log 2$ for both the C and CF scales, and so is multiplying by 2 on both scales, and 2×7 works on CF and DF just fine.

The DF scale has another use, as its starting point of π is fixed directly over the D value of 1, we can get multiples of π by adding log distances just by moving along the D scale. Using the hairline slider over D6 we can view $6 \times \pi = 18.8$ approx by reading it from the DF scale. $(\log \pi + \log 6)$.

If you are reading this just to get a feel of how slide rules work you can probably stop now. However there is a load of magic to come in getting powers and roots, so if this sort of old elegance amuses you, please read on.

3 Powers

Going right back to the section describing logs, we had the result:

$$\log_b y^n = n \log_b y$$

I've used a variable$y$ here, to avoid over-use of x, and, taking logs of both sides:

$$\log_a(\log_b y^n) = \log_a(n \log_b y) = \log_a n + \log_a(\log_b y)$$

Our slide rule uses $\log_{10}$ to multiply two numbers, so this becomes:

$$\log_{10}(\log_b y^n) = \log_{10} n + \log_{10}(\log_b y)$$

On our slide rule, scale D has x inscribed, with $\log_{10} x$ as the distance to the x value.

If x itself was the logarithm (to base b) of another number y, then the distance along the slide rule would be $\log_{10}(\log_b y)$ which is what we need, and gives us a good chance of calculating y^n. So we want another scale, with y inscribed, mapping to the x values, such that:

$$x = \log_b y$$

Or in other words a scale of:

$$b^x$$

So there's a question of how to choose the base b, and there is also an elephant in the room; up to now we have been pretty glib about powers of ten, we can take them or leave them. Alas no longer:

$$2^{0.3} = 1.2311...$$

$$2^3 = 8$$

$$2^{30} = 1073741824$$

Both magnitudes and significant digits are all over the place.

What we do know, is that our D scale has 1.0 at the left end, and 10.0 at the right.

So the left value of our scale b^x will be b and the right hand will be b^{10}.

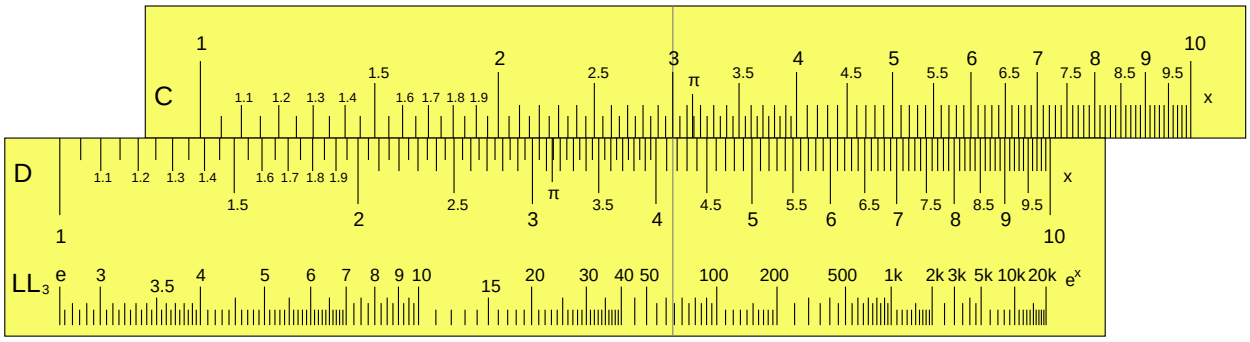
The base of 10 is not very good, it does not have any numbers 1 to 9 on it as it starts at 10, and its maximum value of 10^{10} is only of use to astronomers.

A very common convention, used on most models is to use e, 2.71828... as the base. So the left hand value is 2.718.. and the right hand is e^{10} which is approximately 22026.46, which covers a reasonably practical range of y values.

This has the added facility of an immediate conversion using the hairline slider, from x on the D scale to e^xon our new scale, and also from the y value on our new scale to $\ln y$ on the D scale.

I'll give due warning; we need more than one scale to handle the fact that powers of 10 are significant, usually there are three for y values greater than 1, and three more for y values less than 1, so that's six new scales! At $y = 1$ all powers of y are irrelevant as 1 to any power is just 1. So these multiple scales can get very close to 1, but do not actually get there.

For now we'll only consider the one scale of e^x for the range of x values on the D scale. Here's an illustration, getting 4^3 which is 64:



The 1 of scale C is placed over the 4 of scale LL_3 (using the hairline), then the hairline is moved to C3 and the result, 64 is read on the LL_3 scale.

The breakdown of what is happening:

The 4 of the LL_3 scale maps directly to $\log_e 4$ which is 1.386... on D, then using the C scale to multiply by 3, as we have done before, gives us 4.16.. on D, and this is $\log_e 64$ so we can read 64 directly on the LL_3 scale.

You can also see, just by eye, that the 2 on the C scale lines up with 16 on LL_3, showing 4^2 is sixteen.

The alignment above also shows how roots are taken, the C3 above $LL_3$64 is a division, which is used to calculate $64^{\frac{1}{3}}$ and moving to C1 gets the cubed root on the LL_3 scale which is 4.

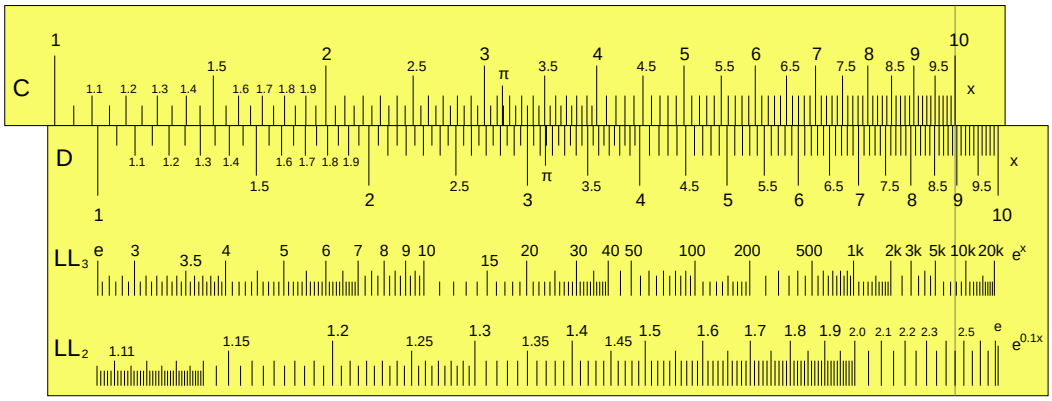
However if we are interested in results which gives us much larger numbers we are running out of luck, the biggest number shown above is 20000, and the resolution of the scale at that end is fairly poor.

You will also see that the starting value of e on LL_3 does not allow powers of smaller numbers, and for example taking the square root of 6, which is 2.45.. is problematic, moving C2 over $LL_3$6 leaves C1 hanging out on the left.

The answer is to add another scale, which is the equivalent of the scale to the left of LL_3.

Remembering that D is from 1 to 10, and if there was a scale to the left of D, it would be from 0.1 to 1. (Those powers of 10!)

Therefore the left-of-LL_3 is $e^{0.1}$ (1.105...) to e (2.718...) and is usually labeled LL_2, here's an illustration:



To obtain the square root of 6, we have placed C2 over $LL_3 6$, and our C1 index has been left hanging over to the left. When this happens we are used to just using the C10 index instead. So on the rule above the hairline has been placed over C10.

In normal multiplication this only leaves us a factor of 10 out, but in this case LL_3 is showing a value around 8000, which is clearly wrong. However we see that the hairline is over LL_2 2.45, which is pretty close to the actual value of 2.449...

So here's the rules:

If you can use the C1 index and read your power and root values on one LL scale, then life is easy.

If you move the C scale left and have to use the C10 index for a root, you go down an LL scale - you are getting a smaller number.

On the other hand if you are getting a power, (for example 2.45^2) and have to use the C10 index, you go up a scale - from LL_2 to $LL_3 6$ in the above illustration, as you are getting a bigger number.

Most slide rules also have an LL_1 scale as well, and some have an LL_0 scale - getting ever closer to 1, but not quite getting there.

3.1 Why that worked

So that's how it works, now a bit more on why. Back to:

$$\log_{10}(\log_e y^n) = \log_{10} n + \log_{10}(\log_e y)$$

Another nice result, if n is a number such as $\frac{1}{2}$ for a square root, the above becomes:

$$\log_{10}(\log_e y^{\frac{1}{n}}) = \log_{10} \frac{1}{n} + \log_{10}(\log_e y)$$

$$\log_{10}(\log_e y^{\frac{1}{n}}) = \log_{10} 1 - \log_{10} n + \log_{10}(\log_e y)$$

But $\log_{10} 1$ is just zero.

$$\log_{10}(\log_e y^{\frac{1}{n}}) = -\log_{10} n + \log_{10}(\log_e y)$$

So raising to a power is adding logs, and getting a root is subtracting logs.

On our illustration above, the $LL_3 6$ value is directly under D 1.792.. which is $\log_e 6$. The distance along the ruler from D1 to this value is $\log_{10}(\log_e 6)$, and as we are looking for a square root we are subtracting $\log_{10} 2$. If we were looking for the square of 6 we would be adding $\log_{10} 2$.

To subtract $\log_{10} 2$ we normally place C2 above $LL_3 6$ and go to the left to C1, and the job is done, but in this case we need to use C10, so we are actually adding a further $\log_{10} 10$, and doing:

$$\log_{10}(\log_e y) - \log_{10} n + \log_{10} 10$$

But we cannot ignore powers of 10, and that value is too big by the addition of $\log_{10} 10$, we are actually getting:

$$\log_{10}(10 \log_e y) - \log_{10} n$$

$$\log_{10}(\log_e y^{10}) - \log_{10} n$$

Putting y=6 and n=2 into this, gives us the square root of 6^{10}, which is 7776, and we have seen that on the LL_3 scale. Hence we need another scale:

$$z = y^{0.1}$$

So we are back to:

$$\log(\log_e z) - \log_{10} n$$

Which we know is simply:

$$\log_{10}(\log_e z^{\frac{1}{n}})$$

and this will give us the correct result on the new scale of z.

And as our LL_3 scale is

$$y = e^x$$

Our new scale LL_2 must be:

$$z = e^{0.1x}$$

Which is indeed what we have, and shows that if we use the C10 index to find a root, we have to drop down to the LL_2 scale to get the answer.

It seems reasonable to think that for multiplication, using the C10 index means going up or down by a factor of 10, whereas on a LogLog scale it means going up or down by a power of 10.

Usually a slide rule will have further scales, of e^{-x}, $e^{-0.1x}$ and $e^{-0.01x}$, these are marked in red, and instead of increasing from left to right, they decrease from left to right. Otherwise they operate in a very similar manner.

The best way to explore this further is to try some calculations on a real Slide Rule, or on one of the web based simulations such as at [2]

In summary, Slide Rules are certainly obsolete, but they have a beauty to those who can appreciate it. If you find your grandparents old Slide Rule in an attic box, don't throw it away, no one will ever make new ones again.

References

[1] https://en.wikipedia.org/wiki/Logarithm

[2] https://sliderules.lovett.com/new_wolf_simulators/webSimulator/index.html